

IDAG-EDIT: MULTI-OBJECT VIDEO EDITING VIA INSTANCE-DECOUPLED ATTENTION AND GUIDANCE

Supplementary Material

1. ADDITIONAL METHOD DETAILS

1.1. Guided Sampling with IA-STDG

We adopt the guided diffusion sampling framework of MotionClone [1] and augment CFG with an auxiliary gradient-based guidance term. During inference, the noise prediction at timestep t is formulated as:

$$\hat{\epsilon}_\theta = \epsilon_\theta(z_t, c, t) + s(\epsilon_\theta(z_t, c, t) - \epsilon_\theta(z_t, \phi, t)) - \lambda\sqrt{1 - \bar{\alpha}_t} \mathcal{G}(t), \quad (1)$$

where $\epsilon_\theta(z_t, \phi, t)$ denotes the unconditional prediction used for CFG, s and λ are guidance weights, and $\bar{\alpha}_t$ is defined by the diffusion noise schedule. Following MotionClone, $\mathcal{G}(t)$ denotes the auxiliary guidance applied during sampling; in our method, it is instantiated by the proposed Instance-Aware Spatial-Temporal Decoupled Guidance (IA-STDG), which refines the denoising trajectory by enforcing instance-consistent spatial-temporal guidance.

1.2. Detail of Modified Text Embedding for Object-Specific Conditioning

Most diffusion based video editing methods rely on a single global caption as a condition, encoding all objects and attributes into a unified contextualized text embedding. In multi-object scenarios, this global conditioning entangles object-specific semantics, causing similarity-based cross-attention to fail in distinguishing target instances, which often leads to attribute leakage, simultaneous editing of multiple objects, or edit failure.

As illustrated in Fig. 1, we mitigate semantic interference between objects directly at the text encoding stage. We first encode a global prompt C and multiple object-specific local prompts using the shared CLIP text encoder¹. We then align the token span of each local prompt with the global sequence and replace the corresponding global embeddings with the local-context embeddings (excluding BOS/EOS tokens). The resulting modified embedding C' serves as the refined text condition for cross-attention during denoising.

¹<https://huggingface.co/openai/clip-vit-large-patch14>

1.3. Implementation Details

Our framework utilizes Stable Diffusion v1.5 [2] integrated with AnimateDiff motion modules [3] as the T2V backbone. Experiments are conducted on a single NVIDIA RTX A6000 GPU. The complete pipeline requires approximately 8 minutes for inversion and 1.5 minutes for editing per video. The full pipeline requires about 8 minutes for inversion and 1.5 minutes for editing. The cross-attention threshold τ_c (Eq. 4) is fixed at 0.8, while the self-attention threshold τ_s (Eq. 8) is adaptively selected in $[0.4, 0.6]$. Following [4], we use $\xi_{\text{cross}}(t) = t^5$ and $\xi_{\text{self}}(t) = 0.3t^5$ for attention bias scheduling, where $t \in [0, 1]$. For guidance weighting (Eq. 3), we set $\eta_f = 0.5$, $\zeta_f = 0$, $\zeta_b = 0.5$, and vary η_b from 0.3 to 0.8 depending on scene complexity and background motion.

2. EXTENDED EXPERIMENTAL ANALYSIS

2.1. Cross Attention Visualization

As illustrated in Fig. 2, our Layout-Guided Cross-Attention (LG-CA) module significantly improves token–subject alignment in complex multi-person scenes. We visualize the cross-attention maps for two distinct prompts: 'Spiderman in red-and-blue suit' and 'Batman in dark armored suit'. Without LG-CA, attention distributions are diffused and exhibit partial overlap between the two subjects, reflecting semantic entanglement where tokens attend to unintended regions. In contrast, applying our layout-aware modulation effectively concentrates the attention of subject and attribute tokens (e.g., 'red', 'armored') onto their respective targets—isolating Spiderman to the left and Batman to the right. This demonstrates that our method successfully mitigates cross-object interference and achieves precise text-to-region correspondence.

2.2. Additional Qualitative Comparisons

As illustrated in Fig. 3, we provide additional qualitative comparisons between IDAG-Edit and recent state-of-the-art methods under three multi-object editing scenarios, where two distinct subjects are required to be transformed into different target identities.

Existing approaches struggle to decouple same-class features within diffusion models. TokenFlow and VideoDirector often exhibit identity coupling, leading to incom-

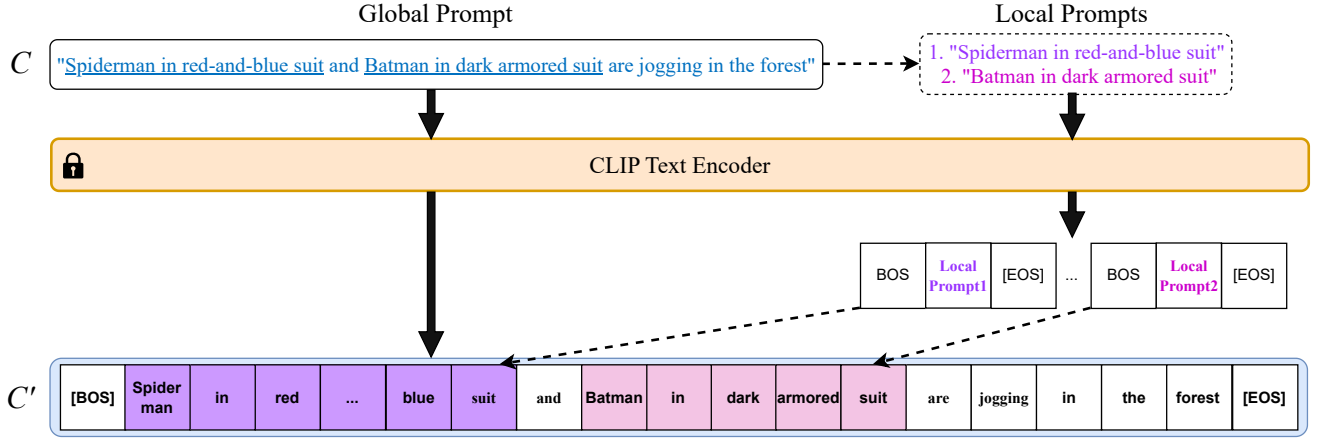


Fig. 1: An illustration of modified text conditioning that replaces global token embeddings with locally encoded object-specific embeddings to reduce cross-object semantic interference in cross-attention.

plete or conflated edits (highlighted with orange boxes). Although Ground-A-Video (GAV) leverages instance-level captions and bounding boxes, it still suffers from attribute mixing and degraded editing fidelity (indicated by blue arrows). VideoGrain, which relies on hierarchical captions and feature-clustered masks, also fails to consistently preserve distinct identities.

Specifically, GAV (3rd row) successfully edits the left subject in the first two examples but fails to correctly transform the right subject into the intended identities (e.g., "right man" into "spiderman" and "batman"). VideoGrain (4th row), when operating without auxiliary modules such as ControlNet, relies solely on masks derived from k-means clustering of spatial features. These masks lack sufficient semantic grounding, resulting in incomplete or failed transformations across all three cases (e.g., missed edits, incorrect identities, or unsuccessful animal conversion). VideoDirector (5th row) further introduces noticeable background artifacts and color distortion, reducing overall realism.

In contrast, IDAG-Edit achieves consistent multi-object transformations without cross-identity interference. The successful editing of multiple subjects can be attributed to our LG-CA, which spatially aligns each object-specific prompt with its designated region. The clear separation of object identities, without texture or color bleeding, reflects the effect of our Instance-Decoupled Self-Attention (ID-SA). Moreover, the stable preservation of fine-grained details across frames indicates the contribution of our IA-STDG. As highlighted in the zoomed-in regions (highlighted with yellow boxes), discriminative attributes—such as Spiderman’s web patterns, the Batman’s emblem, and tabby fur textures—remain sharp and temporally coherent.

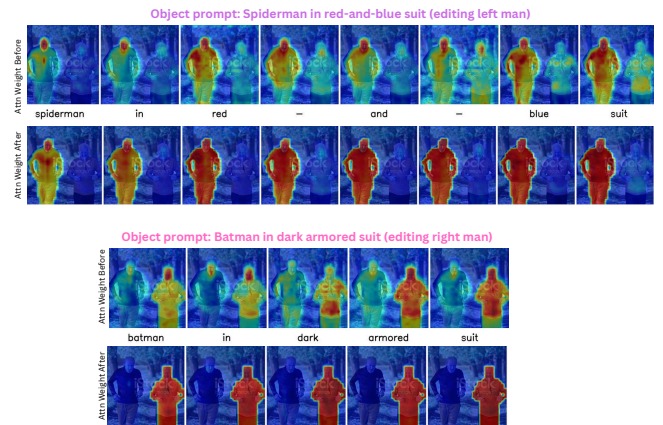


Fig. 2: Attention reweight distribution visualization.



Fig. 3: Additional qualitative comparisons. Unfaithful editing results are highlighted in orange boxes, and instances of attention leakage are indicated by blue arrows. Yellow boxes zoom in on the fine-grained discriminative attributes preserved by our method.

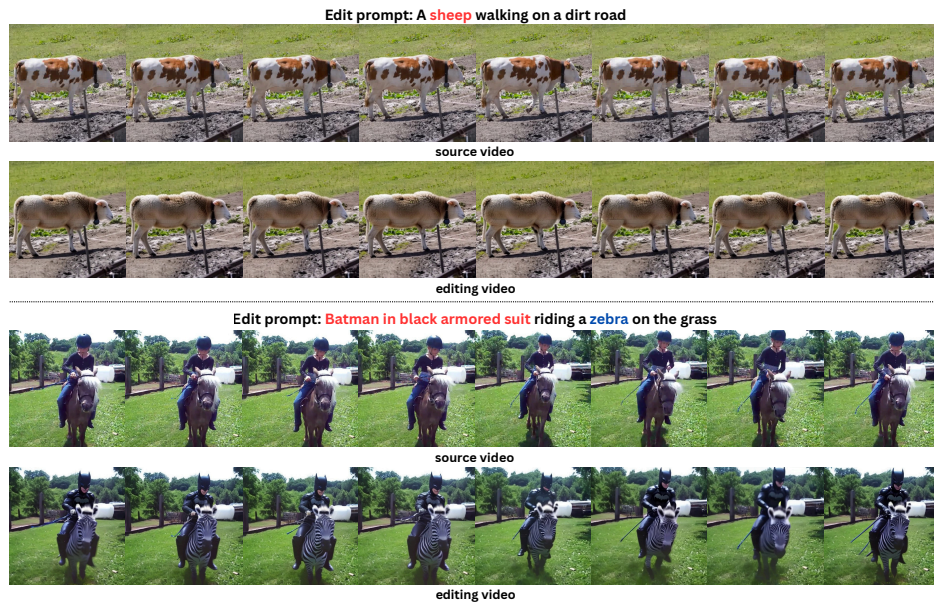


Fig. 4: More general objects instance editing (animals) results.

3. REFERENCES

- [1] P. Ling, J. Bu, P. Zhang, X. Dong, Y. Zang, T. Wu, H. Chen, J. Wang, and Y. Jin, “Motionclone: Training-free motion cloning for controllable video generation,” *arXiv preprint arXiv:2406.05338*, 2024.
- [2] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*. IEEE, 2022, pp. 10 674–10 685.
- [3] Y. Guo, C. Yang, A. Rao, Z. Liang, Y. Wang, Y. Qiao, M. Agrawala, D. Lin, and B. Dai, “Animatediff: Animate your personalized text-to-image diffusion models without specific tuning,” *International Conference on Learning Representations*, 2024.
- [4] Y. Kim, J. Lee, J.-H. Kim, J.-W. Ha, and J.-Y. Zhu, “Dense text-to-image generation with attention modulation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 7701–7711.